

Questions Unit 3

patidar.vinod

October 2025

Questions on Random Number Generators

General Concepts of Random Number Generators

- What are the main differences between True Random Number Generators (TRNGs) and Pseudo-Random Number Generators (PRNGs)?
- Describe the role of a “seed” in Pseudo-Random Number Generators (PRNGs).
- Why is reproducibility important in PRNGs, and in which applications is this feature particularly useful?

Properties of PRNGs

- What does it mean for a PRNG to have “periodicity,” and why is a long period desirable?
- Explain the significance of statistical randomness and non-correlation in the context of PRNGs.
- How does seed sensitivity impact the randomness generated by a PRNG?

Types of PRNGs and Algorithms

- Outline the formula and parameters for a Linear Congruential Generator (LCG).
- How does the Inverse Transform Method generate random samples from a given distribution?
- In the Acceptance-Rejection Method, what are the conditions to accept or reject a sample?

Monte Carlo Simulation Techniques

- What is the Monte Carlo Simulation method, and in which fields is it commonly used?
- Describe the process for estimating π using Monte Carlo Simulation.

- Explain the concept of Monte Carlo Integration and give an example of its application in higher-dimensional integrals.

Advanced Topics and Applications

- How can the Acceptance-Rejection Method be applied to sample from complex probability distributions?
- What is the Box-Muller algorithm, and how does it generate normally distributed random variables?
- Explain the importance of variance reduction techniques, such as importance sampling, in Monte Carlo Integration.

Practical Applications of Monte Carlo Methods

- Describe a scenario where the Monte Carlo method is used to price a European call option.
- How does Monte Carlo Simulation help in estimating integrals for functions without closed-form solutions?

Analytical and Numerical Questions

General Concepts and Properties of PRNGs

- **Analytical:** Explain why true random number generation is often more challenging and expensive compared to pseudo-random generation. Give examples of where TRNGs are typically preferred.
- **Numerical:** Given a PRNG with a seed value of 5 and parameters $a = 1664525$, $c = 1013904223$, and $m = 2^{32}$, use the Linear Congruential Generator formula to calculate the first three numbers in the generated sequence.

Properties of PRNGs

- **Analytical:** Describe the importance of non-correlation in PRNGs and its impact on applications like cryptography.
- **Numerical:** A PRNG generates values between 0 and 1. Given the following generated sequence: 0.23, 0.57, 0.81, 0.12, and 0.44, calculate the mean and variance. Do these values suggest a uniform distribution?

Linear Congruential Generators (LCG) and Multiplicative Congruential Generators (MCG)

- **Analytical:** Compare Linear Congruential Generators (LCG) and Multiplicative Congruential Generators (MCG). Discuss the effect of setting the increment $c = 0$ in MCGs.

- **Numerical:** Given the LCG parameters $a = 16807$, $c = 0$, and $m = 2^{31} - 1$ (Park-Miller LCG), calculate the sequence of numbers starting with seed 10 for the first three iterations.

Inverse Transform Method

- **Analytical:** Explain how the Inverse Transform Method is used to generate samples from an exponential distribution. Why is this method effective for distributions with a well-defined CDF?
- **Numerical:** Generate a sample from an exponential distribution with rate parameter $\lambda = 0.5$ using the Inverse Transform Method. Assume $U = 0.3$ is a uniformly distributed random variable.

Acceptance-Rejection Method

- **Analytical:** Describe how the Acceptance-Rejection Method can be optimized by choosing an appropriate proposal distribution $g(x)$. What factors affect the efficiency of this method?
- **Numerical:** Use the Acceptance-Rejection Method to sample from the distribution $f(x) = \frac{1}{2}e^{-x}$ for $x \geq 0$ with a uniform proposal distribution $g(x) = \frac{1}{4}$ on the interval $[0, 4]$. Given $M = 2$, check if a generated point $x = 1.5$ with $u = 0.3$ is accepted.

Box-Muller Algorithm

- **Analytical:** Describe the transformation performed by the Box-Muller algorithm to generate normally distributed variables. Why does it involve both the sine and cosine functions?
- **Numerical:** Use the Box-Muller algorithm to generate a pair of standard normal random variables from the uniform variables $U_1 = 0.4$ and $U_2 = 0.7$.

Monte Carlo Simulation for Estimation of π

- **Analytical:** Explain the method of estimating π using Monte Carlo simulation and discuss factors that impact the accuracy of the estimate.
- **Numerical:** Suppose you generate 10 random points inside a unit square and find that 8 of these points also fall within a quarter circle inscribed in the square. Estimate the value of π based on this experiment.

Monte Carlo Integration

- **Analytical:** Discuss how Monte Carlo Integration is particularly useful for high-dimensional integrals. Explain the impact of sample size on convergence accuracy.

- **Numerical:** Estimate the integral $\int_0^1 x^2 dx$ using Monte Carlo Integration with 5 sample points: $x_1 = 0.1$, $x_2 = 0.4$, $x_3 = 0.5$, $x_4 = 0.7$, $x_5 = 0.9$. Compare your estimate with the exact value $\frac{1}{3}$.

Monte Carlo Simulation in Option Pricing

- **Analytical:** Outline the steps to use Monte Carlo simulation for pricing a European call option. Discuss how factors like volatility and time to expiration influence the simulation results.
- **Numerical:** For a European call option with stock price $S_0 = 100$, strike price $K = 110$, risk-free rate $r = 0.05$, volatility $\sigma = 0.2$, and time to expiration $T = 1$, simulate the stock price at expiration with $Z = 0.2$ (standard normal variable) and calculate the payoff.

Advanced Monte Carlo Integration Example

- **Analytical:** Discuss the advantages and limitations of using Monte Carlo methods for estimating integrals of non-standard functions, such as $\int_0^1 e^{-x^2} dx$.
- **Numerical:** Estimate the value of the integral $\int_0^1 e^{-x^2} dx$ using Monte Carlo Integration with five sample points: $x_1 = 0.2$, $x_2 = 0.4$, $x_3 = 0.6$, $x_4 = 0.8$, $x_5 = 1.0$.